

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Acceleration (gradient) or equivalent = $\frac{(v-u)}{t}$ (1) Displacement (area) = $ut + \frac{1}{2}t(v-u)$ or $\frac{(u+v)}{2}t$ (1) Substitution and clear algebra seen e.g. $x = ut + \frac{1}{2}t(at)$ seen (1) No reference to graph award a maximum of 2 marks	3			3	3	
	(b)	(i)	$11 \sin 50^\circ [= 8.4 [3] \text{ m s}^{-1}]$		1		1	1	
		(ii)	Attempt at substituting into $H = ut + \frac{1}{2}at^2$ even if incorrect signs (1) $H = 8.4 \times 1.95 - \frac{1}{2} \times 9.81 \times (1.95)^2$ seen (1) Award 2 marks if only this seen $H = -2.27 \text{ [m]}$ (1) accept 2.21 [m] Accept use of $v = 8 \text{ m s}^{-1}$ leading to $H = 3.05 \text{ [m]}$. Alternative solution: Time to reach max height = $\frac{(0-8.4)}{-9.81}$ (or 0.86 s) (1) Time from H to floor = $1.95 - (2 \times 0.86)$ (or 0.232 s) (1) $H = (8.4 \times 0.23) + \frac{1}{2} \times 9.81 \times (0.23)^2$ (or 2.21 m) (1) Consider carefully consequence of 'rounding'	1 1	1		3	3	
		(iii)	Horizontal component of velocity = $11 \cos 50^\circ [= 7.1 \text{ m s}^{-1}]$ or by Pythagoras (1) Substitution into $v_H = \frac{R}{t}$ $R = 7.1 \text{ (ecf)} \times 1.95 = 13.8 \text{ [m]}$ (1)		2			2	

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	(c)		The time for which the shot is in the air will increase (Steve correct here - no explanation needed) (1) However, increasing the angle will decrease the horizontal component of velocity (1) Reasonable conclusion based on one or both factors e.g. both of these factors will tend to cancel each other out, so not a good idea to follow Steve's advice or it is a bad idea because the angle needs to be decreased (1)			3	3		
			Question 6 total	5	4	3	12	9	0